

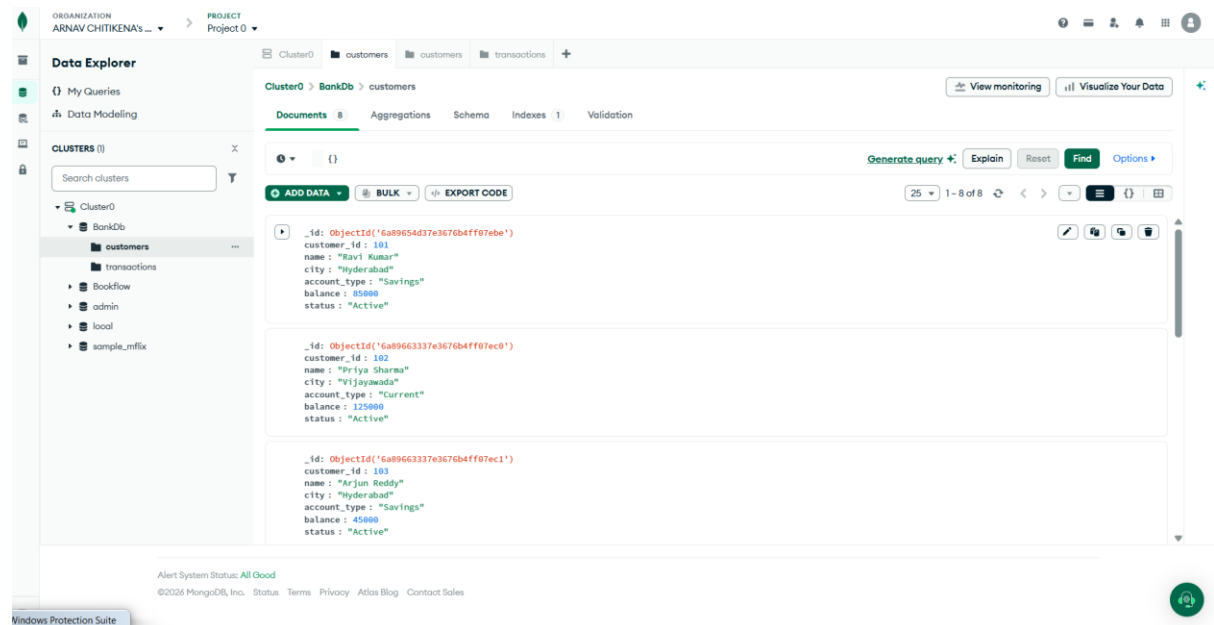
DBSE & DBD SKILL SESSION

Arnav. Chitikena 2520030331

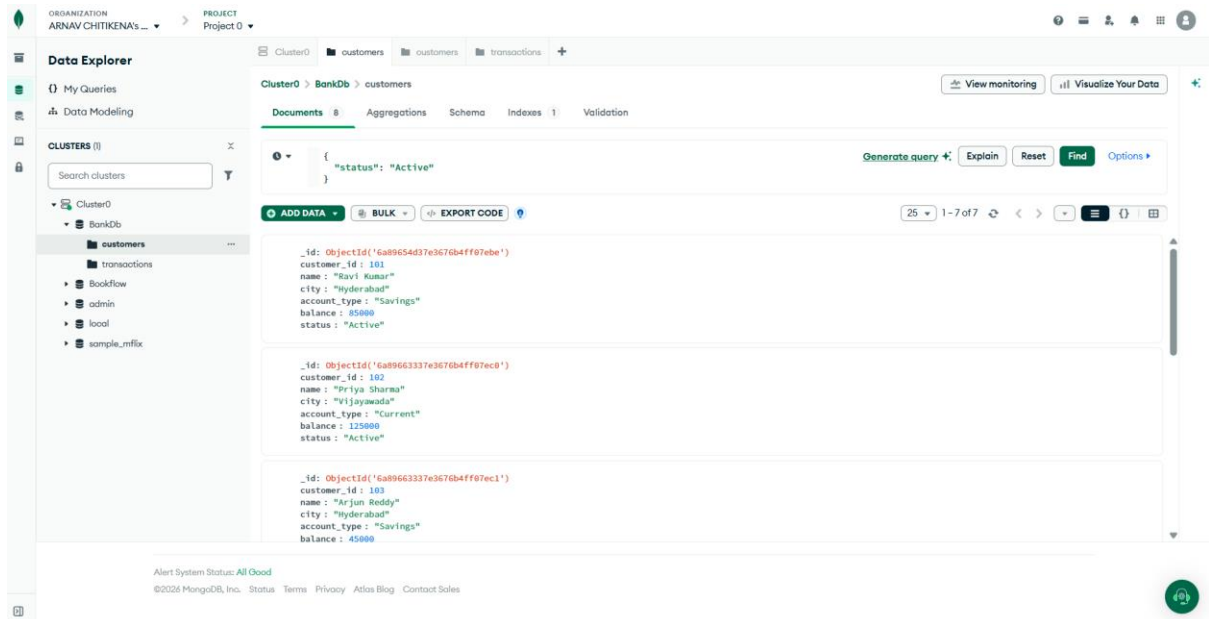
EXPERIMENT TITLE:

Bank Transaction Management Using MongoDB Atlas Data Explorer

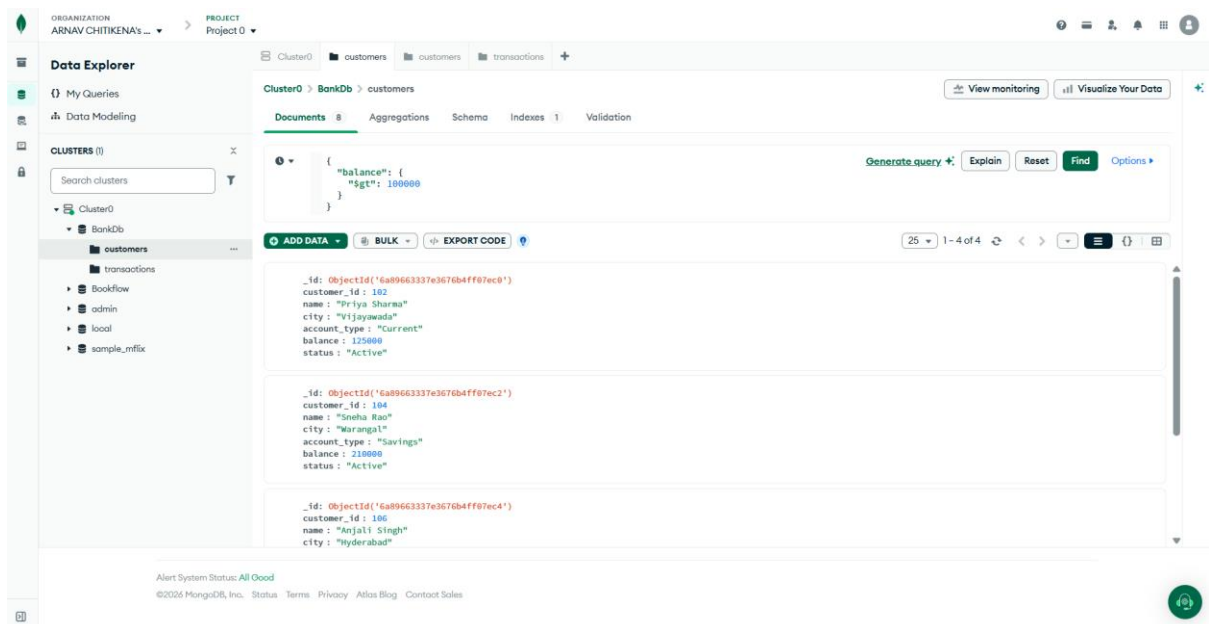
QUERY 1: DISPLAY ALL CUSTOMERS



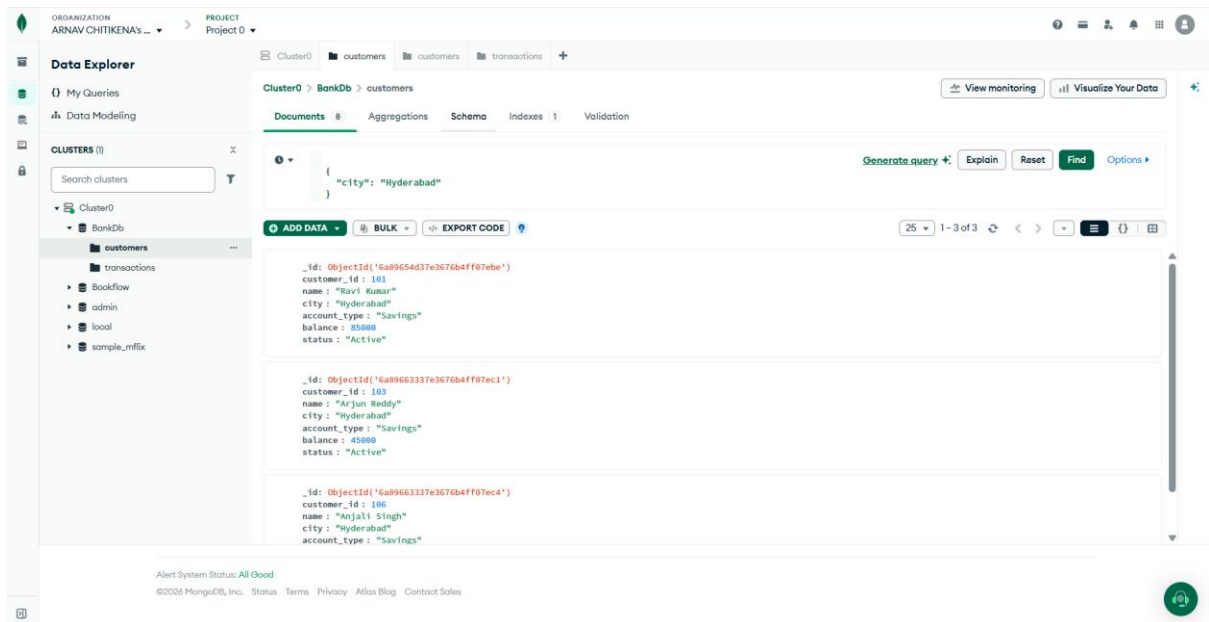
QUERY 2: FIND ACTIVE CUSTOMERS



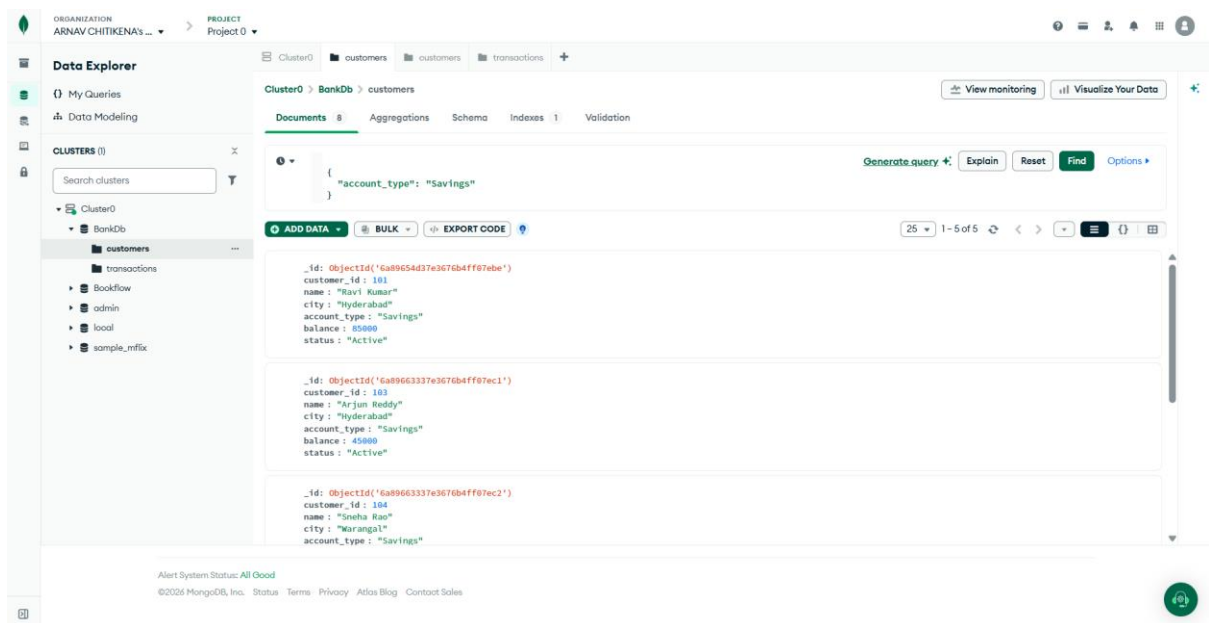
QUERY 3: FIND CUSTOMERS WITH BALANCE GREATER THAN 100000



QUERY 4: FIND HYDERABAD CUSTOMERS



QUERY 5: FIND SAVINGS ACCOUNT CUSTOMERS



QUERY 6: FIND TRANSACTIONS GREATER THAN 20000

ORGANIZATION: ARNAV CHITIKENA'S ... PROJECT: Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

BankDb

customers

transactions

Bookflow

admin

local

sample_mflix

Cluster0 > BankDb > transactions

Documents 15 Aggregations Schema Indexes 1 Validation

Generate query Explain Reset Find Options

```
{
  "amount": {
    "$gt": 20000
  }
}
```

ADD DATA BULK EXPORT CODE

25 1-9 of 9

```
{
  "_id": ObjectId('6a8967fc37e3676b4ff07ec9'),
  "transaction_id": 1001,
  "customer_id": 101,
  "type": "Deposit",
  "amount": 25000,
  "mode": "UPI",
  "date": "2026-08-01",
  "status": "Success"
}
```

```
{
  "_id": ObjectId('6a8967fc37e3676b4ff07ecb'),
  "transaction_id": 1003,
  "customer_id": 102,
  "type": "Deposit",
  "amount": 50000,
  "mode": "NEFT",
  "date": "2026-08-02",
  "status": "Success"
}
```

```
{
  "_id": ObjectId('6a8967fc37e3676b4ff07ecd')
}
```

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QUERY 7: FIND DEPOSITS

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Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

BankDb

customers

transactions

Bookflow

admin

local

sample_mflix

Cluster0 > BankDb > transactions

Documents 15 Aggregations Schema Indexes 1 Validation

Generate query Explain Reset Find Options

```
{
  "type": "Deposit"
}
```

ADD DATA BULK EXPORT CODE

25 1-8 of 8

```
{
  "_id": ObjectId('6a8967fc37e3676b4ff07ec9'),
  "transaction_id": 1001,
  "customer_id": 101,
  "type": "Deposit",
  "amount": 25000,
  "mode": "UPI",
  "date": "2026-08-01",
  "status": "Success"
}
```

```
{
  "_id": ObjectId('6a8967fc37e3676b4ff07ecb'),
  "transaction_id": 1003,
  "customer_id": 102,
  "type": "Deposit",
  "amount": 50000,
  "mode": "NEFT",
  "date": "2026-08-02",
  "status": "Success"
}
```

```
{
  "_id": ObjectId('6a8967fc37e3676b4ff07ecd'),
  "transaction_id": 1005,
  "customer_id": 103
}
```

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QUERY 8: FIND ATM TRANSACTIONS

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Data Explorer

My Queries | Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - BankDb
 - customers
 - transactions

Cluster0 > BankDb > transactions

Documents 15 | Aggregations | Schema | Indexes 1 | Validation

Generate query | Explain | Reset | Find | Options

25 | 1 - 7 of 7

```
{
  "mode": "ATM"
}
```

ADD DATA | BULK | EXPORT CODE

Transaction 1:

```
{
  "_id": ObjectId('6a8967fc37e3676b4ff07eca'),
  "transaction_id": 1892,
  "customer_id": 101,
  "type": "Withdrawal",
  "amount": 10000,
  "mode": "ATM",
  "date": "2026-08-02",
  "status": "Success"
}
```

Transaction 2:

```
{
  "_id": ObjectId('6a8967fc37e3676b4ff07ecc'),
  "transaction_id": 1804,
  "customer_id": 102,
  "type": "Withdrawal",
  "amount": 15000,
  "mode": "ATM",
  "date": "2026-08-03",
  "status": "Success"
}
```

Transaction 3:

```
{
  "_id": ObjectId('6a8967fc37e3676b4ff07ece'),
  "transaction_id": 1806,
  "customer_id": 103
}
```

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QUERY 9: FIND DEPOSITS GREATER THAN 50000

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Data Explorer

My Queries | Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - BankDb
 - customers
 - transactions

Cluster0 > BankDb > transactions

Documents 15 | Aggregations | Schema | Indexes 1 | Validation

Generate query | Explain | Reset | Find | Options

25 | 1 - 3 of 3

```
{
  "type": "Deposit",
  "amount": {
    "$gt": 50000
  }
}
```

ADD DATA | BULK | EXPORT CODE

Transaction 1:

```
{
  "_id": ObjectId('6a8967fc37e3676b4ff07ecf'),
  "transaction_id": 1907,
  "customer_id": 104,
  "type": "Deposit",
  "amount": 100000,
  "mode": "NET",
  "date": "2026-08-04",
  "status": "Success"
}
```

Transaction 2:

```
{
  "_id": ObjectId('6a8967fc37e3676b4ff07ed2'),
  "transaction_id": 1910,
  "customer_id": 105,
  "type": "Deposit",
  "amount": 75000,
  "mode": "NET",
  "date": "2026-08-06",
  "status": "Success"
}
```

Transaction 3:

```
{
  "_id": ObjectId('6a8967fc37e3676b4ff07ed6')
}
```

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QUERY 10: FIND HYDERABAD CUSTOMERS WITH HIGH BALANCE

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Data Explorer

My Queries | Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - BankDb
 - customers
 - transactions
 - Bookflow
 - admin
 - local
 - sample_mflix

Cluster0 > BankDb > customers

Documents | Aggregations | Schema | Indexes | Validation

Generate query | Explain | Reset | Find | Options

```
{
  "city": "Hyderabad",
  "balance": {
    "sgt": 100000
  }
}
```

ADD DATA | BULK | EXPORT CODE

25 | 1-1 of 1

```
{
  "_id": ObjectId('6a8966337e3676b4ff67ec4')
  customer_id: 100
  name: "Anjali Singh"
  city: "Hyderabad"
  account_type: "savings"
  balance: 175000
  status: "Active"
}
```

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QUERY 11: FIND CUSTOMERS IN MULTIPLE CITIES

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Data Explorer

My Queries | Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - BankDb
 - customers
 - transactions
 - Bookflow
 - admin
 - local
 - sample_mflix

Cluster0 > BankDb > customers

Documents | Aggregations | Schema | Indexes | Validation

Generate query | Explain | Reset | Find | Options

```
{
  "city": {
    "$in": [
      "Hyderabad",
      "Vijayawada",
      "Warangal"
    ]
  }
}
```

ADD DATA | BULK | EXPORT CODE

25 | 1-5 of 5

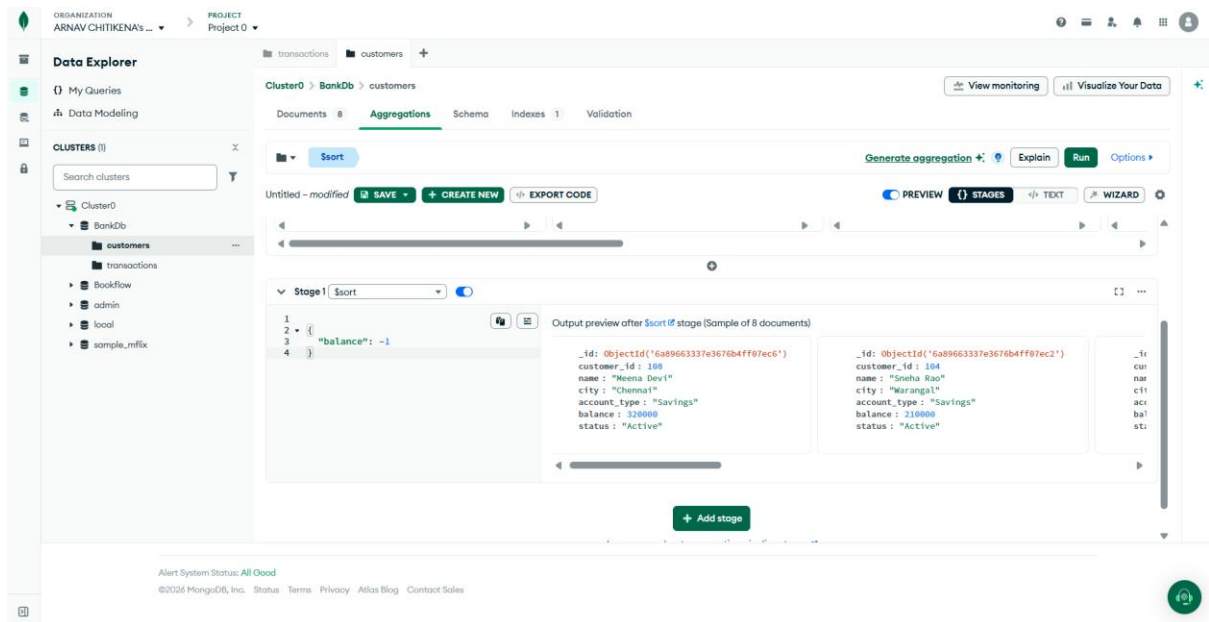
```
{
  "_id": ObjectId('6a89654837e3676b4ff67ec9')
  customer_id: 101
  name: "Ravi Kumar"
  city: "Hyderabad"
  account_type: "Savings"
  balance: 50000
  status: "Active"
}
```

```
{
  "_id": ObjectId('6a8966337e3676b4ff67ec9')
  customer_id: 102
  name: "Priya Sharma"
  city: "Vijayawada"
  account_type: "Current"
  balance: 120000
  status: "Active"
}
```

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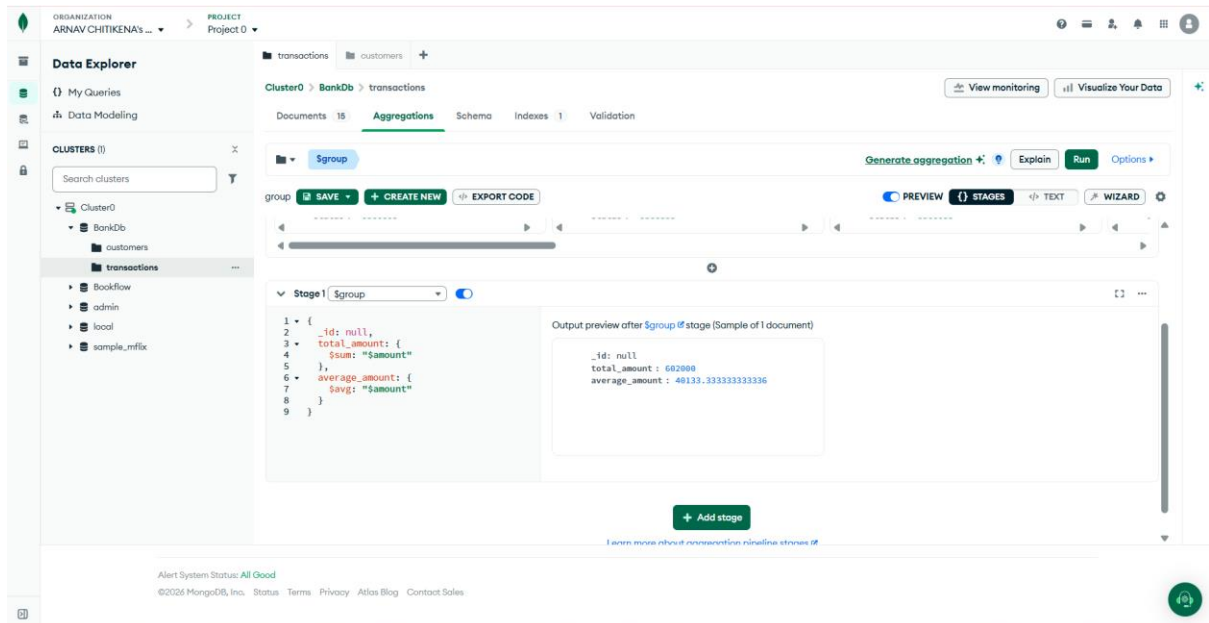
QUERY 12: SORT CUSTOMERS BY BALANCE



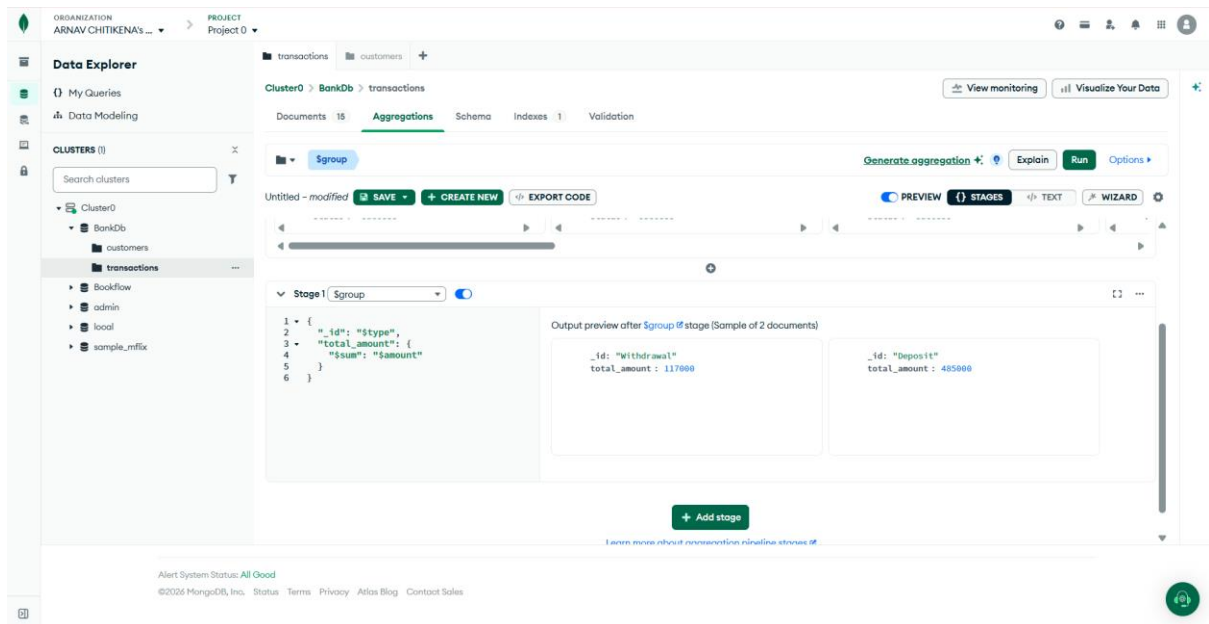
QUERY 13: CALCULATE TOTAL TRANSACTION AMOUNT

QUERY 14: CALCULATE AVERAGE TRANSACTION AMOUNT

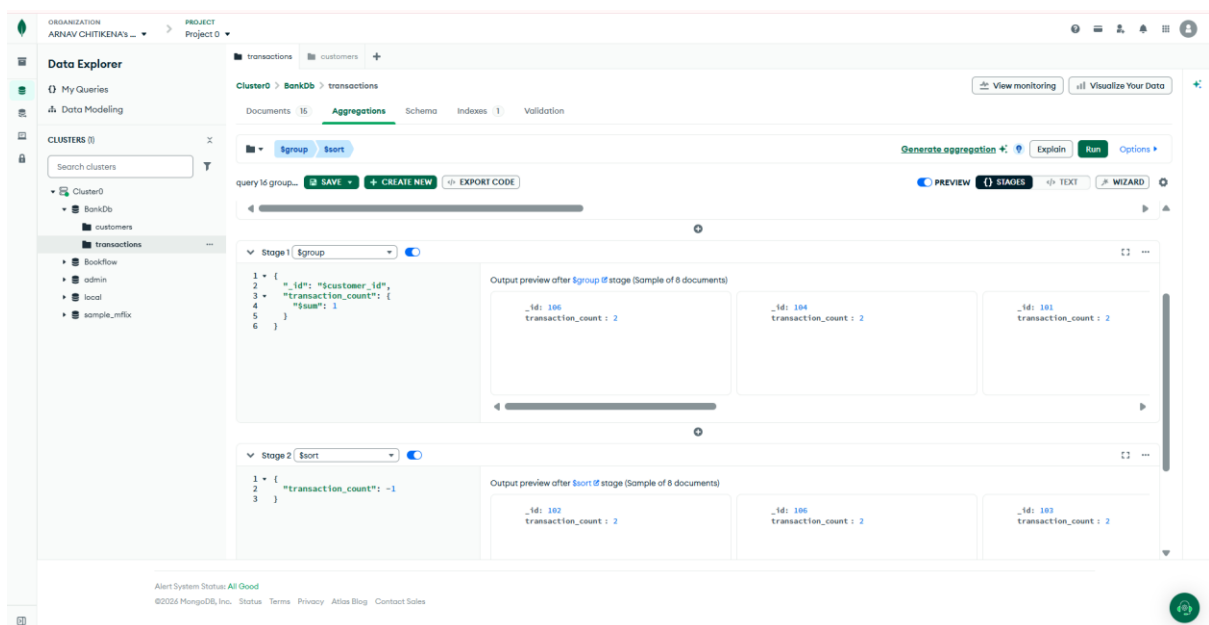
BOTH ARE IN SAME SCREENSHOT



QUERY 15: TOTAL TRANSACTION AMOUNT BY TYPE



QUERY 16: NUMBER OF TRANSACTIONS PER CUSTOMER



ORGANIZATION
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PROJECT
Project 0

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- BankDb
 - customers
 - transactions
- Bookflow
- admin
- local
- sample_mflix

transactionscustomers+

Cluster0 > BankDb > transactions

DocumentsISAggregationsSchemaIndexes1Validation

\$group

\$sort

Generate aggregation+

Explain

Run

Options

query id group...

SAVE

+ CREATE NEW

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

Stage 1 \$group

1 {
2 "customer_id",
3 "transaction_count": {
4 "\$sum": 1
5 }
6 }

Output preview after \$group if stage (Sample of 8 documents)

_id: 185 transaction_count: 1	_id: 188 transaction_count: 2	_id: 189 transaction_count: 2
----------------------------------	----------------------------------	----------------------------------

Stage 2 \$sort

1 {
2 "transaction_count": -1
3 }

Output preview after \$sort if stage (Sample of 8 documents)

_id: 181 transaction_count: 2	_id: 187 transaction_count: 2	_id: 184 transaction_count: 2
----------------------------------	----------------------------------	----------------------------------

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